NATO Communications and Information Agency (NCIA)

DATA CHALLENGE INSTRUCTIONS

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1. Introduction

1.1 Background

NATO has significantly increased its focus on protecting personnel, assets, and missions against the misuse of Class I Unmanned Aircraft Systems (UAS), including small, commercially available drones.

Over recent years, NCIA has conducted extensive experimentation and testing in the C-UAS domain, including large-scale NATO Technical Interoperability Exercises (TIEs) and targeted trials. These activities involved NATO nations, partner nations, and industry interoperability using recognized standards such as SAPIENT, Link 26, and ASTERIX.

These exercises generated a unique, real-world dataset covering a wide range of operational environments, sensor types, and threat scenarios. The Data Challenge builds directly on this experience and data.

1.2 Objectives

The objective of the C-UAS Data Challenge is to identify and evaluate innovative technologies capable of fusing data from multiple sensor types, including: Radar; Radio Frequency (RF) sensors; Electro-Optical / Infrared (EO/IR) sensors; Acoustic sensors; Cyber and network-based sensors.

Participants are expected to develop an AI-enabled innovative, data-driven solution to track, classify and identify Class I UAS, showing technical feasibility, operational relevance to the C-UAS domain, and potential for real-world deployment. The proposed technique should:

- Eliminate duplicate tracks reported by multiple sensors;
- Fuse kinematic and attribute data to improve track quality;
- Classify detected objects (e.g. drone, bird, specific drone type);
- Identify tracks as friendly, hostile, or unknown (e.g. based on white-listed drone identifiers).

The outcome should be a single, coherent recognised air picture suitable for operational use.

1.3 Prize money & award event

The C-UAS Data Challenge have a total prize fund of €100,000, distributed across three places. Final rankings will be confirmed ahead of the awards ceremony in February 2027, when winning teams will present their solutions and receive their prizes (Table (1)).

Table 1 Winners prizes and announcement location

Place	Prize	Event
1st Place	€45,000	*Location tbd*
2nd Place	€35,000,	*Location tbd*
3rd Place	€20,000	*Location tbd*
Total Prize Fund	**€100,000**	*Location tbd*

1.4 Participants eligibility

The competition is open to entities and individuals from all NATO nations and NATO nations only, which includes universities, academia, research institutions and industries.

In order to be eligible (Section 1.4.1.), participants must be sponsored by their corresponding university, academia, research institution or industry. Individuals without a sponsorship are still eligible to participate in the competition and will to the final ranking only (Section 1.5).

1.4.1 Prize money award eligibility

Participation in the Competition is open to all eligible participants. However, prize awards are limited to eligible commercial entities and not-for-profit organisations. Individuals and other participants who do not meet the prize eligibility requirements are welcome to participate for ranking purposes but will not be eligible to receive a prize.

The entity types which can be awarded are:

- *Commercial Supplier*, i.e. all businesses which engage in any form of for-profit-based activities such as selling supplies or services in one or more countries. This category includes, but is not limited to: individual proprietorship (sole traders), partnerships, limited liability companies or corporations.
- *Non-for-Profit Supplier*, i.e. all not-for-profit organizations which engage in any form of not-for-profit-based activities in one or more countries. This category includes, but is not limited to: national defence labs, universities, and non-governmental and other research institutions.

1.5 Competition participation

Participation in the Competition is open to all eligible participants.

The Competition will maintain two rankings:

I. *Overall Ranking,* comprising all participants and reflecting the performance of all valid submissions;
II. *Prize-Eligible Ranking,* comprising only participants that meet the applicable prize eligibility requirements.

The 1st, 2nd and 3rd place prizes will be awarded based on the Prize-Eligible Ranking, irrespective of the positions of those participants in the Overall Ranking. Consequently, a participant may achieve a high position, including first place, in the Overall Ranking without being eligible for a prize.

By entering or otherwise participating in this Competition, participants acknowledge and agree to these Competition Terms, including the eligibility requirements, ranking methodology and conditions applicable to the award of prizes. Participation, submission of an entry, or achievement of a particular position in the Overall Ranking does not in itself establish an entitlement to a prize. Any prize award is subject to verification of the participant's eligibility and compliance with the applicable prize award conditions.

2. Competition instructions

This section presents the instructions that shall be followed by the participants for the data challenge competition.

2.1 Participants registration

Under the condition reported in Section 1.4, participants have the possibility to apply individually or as a group.

After acceptance of registration, each participant will form a team which will run the competition. Each team has its own secured, isolated dedicated folder which can be access from the tab “Upload Team Submission” of the extranet portal available at the following link: https://extranet.ncia.nato.int/CUAS-DataChallenge/.

Each team folder:

- Is accessible by the participants/groups part of the same specific team and the competition organizers only;
- Supports secure file upload/download;
- Should be used by participants to upload their proposed solution results only.

Please note the following to access the SharePoint portal:

- If the above provided link does not work, please access by using the “Extranet portal” option in the left-side panel of the Webex portal;
- You will be redirected to a page asking for username and password:

    1. Use the email you registered with as username;
    2. Click on the “Forgot Password” option and set up a new password.

Please take into consideration that a “*Sorry, this site has not been shared with you*” error may occur, even after re-setting the password. If so, please close the tab and re-lunch the link in a new tab. If the issue persist, please contact the C-UAS Data Challenge Team (data.challenge@ncia.nato.int).

2.2 General rules

The competition is organized in three main phases:

I. *Phase 1: Training*.
All teams must test their proposed solution by using the corresponding set of data as reported in Section 2.1.

II. *Phase 2: Evaluation*.
All teams must submit their proposed solution results obtained from the corresponding set of data as reported in Section 2.1. Results will be evaluated and five (5) best solutions will be selected.

III. *Phase 3: Post-Evaluation*.
Teams of the selected five (5) best solutions must separately attend a one (1) hour online board session (45 min (1.,2.) + 15 min (3.)) during which they should:

3. Present the methodology, data approach and description of their proposed solution;
4. Make a live demonstration by applying their proposed solution to a new corresponding set of data shared with the five (5) best solution participants only at the time of the board session;
5. Reply to board panel questions.

Results will be evaluated and the three (3) best winning solution will be selected.

NCIA provides the teams with curated and anonymized datasets derived from NATO exercises. Results obtained from the specific set of data should be submitted by following a specific submission template shared with each team.

The proposed solutions results are evaluated against objective performance criteria for tracking accuracy, classification performance, identification reliability and sensor fusion completeness. The team with the highest score will be chosen as winner.

2.3 Dataset structure

All teams shall use the dataset folders in the extranet portal available at the following link:

https://extranet.ncia.nato.int/CUAS-DataChallenge/Shared%20Documents/Forms/AllItems.aspx

The proposed dataset is organized in two main folders including sensor data from multiple sensor types as well as reference data. Sensor data and reference data detailed structures description are included in the corresponding README files which can be found in the extranet portal at the following link: https://extranet.ncia.nato.int/CUAS-DataChallenge/ChallengeLibrary/README.
The two datasets folders are organized as reported in Table 2 and they are structured as follows.

I. *Testing datasets.*
This dataset is provided to test the solution and/or train the proposed models and it must be used with the only purpose of:

- Label, test or validate the proposed solution for UAS classification and identification;
- Test or validate results of the proposed solution for UAS tracking.

The training dataset contains eight subfolders, each corresponding to a specific scenario and iteration. Each subfolder comprises:

- Ground truth data consisting of Dronetag files acquired for each flight of each UAS flying during the specific iteration of the specific scenario;
- Sensor data acquired for multiple sensors operating during the particular scenario/iteration.
- Result template consisting of the .xlsx submission file template to be populated with the team' proposed solution test results.

II. *Evaluation datasets.*
This dataset is provided to submit the proposed solution results and it must be used with this purpose only.
The evaluation dataset contains three subfolders, each corresponding to a specific scenario and iteration. Each subfolder comprises:

- List of friendly (i.e. Blue) UASs flying during the specific iteration of the specific scenario.
- Sensor data acquired by multiple sensors operating during the specific iteration of the specific scenario.
- Result template consisting of the .xlsx submission file template to be populated with the team' proposed solution results.

Please note that the provided datasets must be used for the scope of this competition and this competition only. If you would like to use these data for other purposes we kindly ask you to contact the organizers (data.challenge@ncia.nato.int).

Table 2 Datasets folders structure

Testing Dataset			*Evaluation Dataset (Phase 1)*		
Scenario 1	Iteration 1	Sensor data	Scenario 1	Iteration 3	Sensor data
		Ground truth			Friend drone list
Scenario 2	Iteration 2	Sensor data	Scenario 2	Iteration 1	Sensor data
		Ground truth			Friend drone list
Scenario 3	Iteration 1	Sensor data	Scenario 2	Iteration 3	Sensor data
		Ground truth			Friend drone list
	Iteration 2	Sensor data	*Evaluation Dataset (Phase 2)*		
		Ground truth	Scenario 1	Iteration 2	Sensor data
	Iteration 3	Sensor data			Friend drone list
		Ground truth			
Scenario 4	Iteration 4	Sensor data			
		Ground truth			

2.4 Results and Submission Requirements

2.4.1 Challenge Statements

The teams proposed solution should provide results for each scenario and iteration set of data in the Evaluation Datasets. Results should be provided as follows:

I. *Track of Class I UAS.*
The results should provide specifications of the detected objects tracks by reporting information on the corresponding date, time and location.

II. *Classification of Class I UAS.*
The results should provide specifications of the detected objects types by reporting the corresponding classification information.

III. *Identification of Class I UAS.*
The results should provide specifications of the detected objects identity.

IV. *Fusion Completeness*
The results should provide specifications of the sensors used to form the objects fused tracks.

Teams must report their proposed solution results in a .xlsx file format following the submission templates provided for each scenario and iteration set of data in the Evaluation Datasets. Specific results information required as well as instruction on how to populate the .xlsx submission templates are included in the "README" file which can be found in the extranet portal at the following link: https://extranet.ncia.nato.int/CUAS-DataChallenge/ChallengeLibrary/README.

2.4.2 Submission Requirements

Teams should submit their results and supporting documentation in their specific team folder on the extranet portal (https://extranet.ncia.nato.int/CUAS-DataChallenge/SitePages/Home.aspx).

Procedures for the teams' results submission are reported below.

- Teams must re-name the results .xlsx file as reported in the README file available at this link https://extranet.ncia.nato.int/CUAS-DataChallenge/ChallengeLibrary/README/README_SubmissionTemplate.pdf ;
- Teams must upload the results .xlsx file in the '*Submission'* folder of their specific team folder;
- Before the competition closing date, teams have the possibility to upload multiple results for the same set of data;
- Once uploaded, the results .xlsx file is periodically stored in a specific folder automatically created for the corresponding uploading-date/time and named accordingly as '*YYYMMDDhhmm*'. PLEASE NOTE: Unless teams would like to overwrite the uploaded results, they must wait until their submitted result file is moved to the corresponding folder before submitting a new result file.

Furthermore, teams should note the following:

- The selected five (5) best solution teams must submit documentation reporting a clear methodology, data approach and description of the AI/ML techniques applied as .pdf file in the *'Methodology and prototype documentation'* folder of their specific team folder;
- The three (3) best solution teams must submit supporting evidence or prototypes to demonstrate feasibility, where applicable, in the *'Methodology and prototype documentation'* folder of their specific team folder.

Submissions of the teams' proposed solution is subject to review and inspection by the challenge organisers which will not be made open source. Intellectual Property Rights (IPR) within challenge submissions shall remain with the team.

2.5 Evaluation and Scoring procedures

Procedures for the teams submitted results evaluation and scoring are reported below.

- The submitted results are periodically evaluated and scored, after they have been uploaded in the corresponding folder of the specific team folder;
  Once submissions start to be received and the evaluation process begins, each scoring result is published in the *'Leaderboard"* folder on the extranet portal at the following link: https://extranet.ncia.nato.int/CUAS-DataChallenge/Shared%20Documents/Leaderboards.
  PLEASE NOTE:
  - One leaderboard per each set of data is displayed and periodically updated to give a real-time view of how the teams' time compares;
  - Live standings for all teams can be also accessed from the *Leaderboard Tab* of the Conference Hub;
- After the competition closing date, per each set of data the results which obtained the highest score is chosen;
- After comparison between teams, teams with the reported highest scores are chosen as winners.
- Final rankings will be confirmed ahead of the awards ceremony in November 2026.

Teams' submitted results are evaluated by comparison with the corresponding ground-truth data against the performance criteria reported below.

- *Track of Class I UAS performance.*
  The multi-object tracking performance of the participant proposed solution is evaluated by the GOSPA method.

- *Classification of Class I UAS accuracy.*
  The classification accuracy of the participant proposed solution is evaluated by the Recall metric.

- *Identification of Class I UAS accuracy.*
  The identification accuracy of the participant proposed solution is evaluated by the Recall metric.

- *Fusion of sensor data.*
  The sensor data fusion completeness of the participant proposed solution is evaluated by measuring the reported class metadata correctness.

The combined scored of the participants' result is calculated from the obtained evaluation values weighted as reported in Table 3.

Table 3 Scores weight values

Evaluation areas	*Weight*
Track of Class I UAS accuracy	0.4
Correct Classification of Class I UAS	0.2
Correct Identification of Class I UAS	0.1
Fusion Completeness	0.3